

Some of the Popular Laser Types and Their Applications

Laser	Pump source	Operation wavelengths	Applications
Gas lasers			
Helium-Neon laser	Electrical discharge	632.8 nm (543.5 nm, 593.9 nm, 611.8 nm, 1.1523 μm , 1.52 μm , 3.3913 μm)	Interferometry, holography, spectroscopy
Carbon-dioxide laser	Transverse or longitudinal electrical discharge	10.6 μm (9.4 μm)	Material processing (cutting, welding, etc), surgery
Argon-ion laser	Electrical discharge	454.6 nm, 488.0 nm, 514.5 nm (351 nm, 363.8, 457.9 nm, 465.8 nm, 476.5 nm, 472.7 nm, 528.7 nm, also frequency doubled to provide 244 nm, 257 nm)	Retinal phototherapy (for diabetes), lithography, spectroscopy, pumping other lasers
Krypton-ion laser	Electrical discharge	416 nm, 530.9 nm, 568.2 nm, 647.1 nm, 676.4 nm, 752.5 nm, 799.3 nm	Scientific research, mixed with argon to create white-light lasers
Helium-cadmium metal vapour laser	Electrical discharge in metal vapour mixed with helium buffer gas	441.563 nm, 325 nm	Printing and typesetting
Copper vapour laser	Electrical discharge	510.6 nm, 578.2 nm	Dermatological uses, high-speed photography
Chemical lasers			
Hydrogen fluoride laser	Chemical reaction in a burning jet of ethylene and nitrogen trifluoride (NF ₃)	2.7 to 2.9 μm for hydrogen fluoride (<80 per cent atmospheric transmittance)	Used in research for laser weaponry by the US DOD
Deuterium fluoride laser	Chemical reaction	3.6 μm to 4.2 μm (90 per cent atmospheric transmittance)	MIRACL, tactical high-energy laser
Chemical oxygen-iodine laser	Chemical reaction in a jet of singlet delta oxygen and iodine	1.315 μm (<70 per cent atmospheric transmittance)	Laser weaponry, scientific and materials research, laser used in the US military's airborne laser
All gas-phase iodine laser	Chemical reaction of chlorine atoms with gaseous hydrazoic acid	1.315 μm (<70 per cent atmospheric transmittance)	Scientific, weaponry, aerospace
Dye lasers			
Dye laser	Other laser	390-435 nm, 460-515 nm, 570-640 nm	Research, laser medicine, spectroscopy
Solidstate lasers			
Ruby laser	Flash lamp	694.3 nm	Holography, tattoo removal
Er: YAG laser	Flash lamp, laser diode	2.94 μm	Periodontal scaling, dentistry
Nd: YAG laser	Flash lamp, laser diode	1.064 μm (1.32 μm)	Material processing, range finding, laser target designation
Nd: YLF solidstate laser	Flash lamp, laser diode	1.047 and 1.053 μm	Mostly used for pulsed pumping of certain types of pulsed lasers
Nd: Glass laser	Flash lamp, laser diode	1.062 μm (silicate glasses), 1.054 μm (phosphate glasses)	Used in extremely high-power, high-energy multiple-beam systems for inertial confinement fusion, usually frequency tripled to the third harmonic at 351 nm in laser fusion devices
Neodymium-doped yttrium orthovanadate Nd: YVO ₄ laser	Laser diode	1.064 μm	Mostly used for continuous pumping of Ti:sapphire or dye lasers
Titanium sapphire (Ti:sapphire) laser	Other laser	650-1100 nm	Spectroscopy, LIDAR, Often used in highly-tunable mode-locked infrared lasers to produce ultra-short pulses and in amplifier lasers to produce ultra-short and ultra-intense pulses
Semiconductor laser			
AlGaAs	Electrical current	0.63-0.9 μm	Optical disks, laser pointers, data communications
InGaAsP	Electrical current	1.0-2.1 μm	Telecommunications, machining
GaN	Electrical current	0.4 μm	Optical disks
Other lasers			
Free-electron laser	Relativistic electron beam	A broad wavelength range (about 100 nm to several nm)	Atmospheric research, material science, medical applications
Gas dynamic laser	Spin state population inversion in carbon dioxide molecules caused by supersonic adiabatic expansion of mixture of nitrogen and carbon dioxide	Several lines around 10.5 μm	Military applications